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This program is awaiting a new solicitation and is not currently accepting proposals.

NSF 25-542: Smart Health and Biomedical Research in the Era of Artificial Intelligence and Advanced Data Science (SCH)

Program Solicitation

Document Information

Document History

- **Posted:** July 24, 2025
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U.S. National Science Foundation

Directorate for Computer and Information Science and Engineering

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Division of Computing and Communication Foundations

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National Institute of Arthritis and Musculoskeletal and Skin Disease

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National Institute of Nursing Research
National Library of Medicine
National Heart, Lung, and Blood Institute
National Institute on Minority Health and Health Disparities
National Center for Advancing Translational Sciences
Office of Disease Prevention
Office of Nutrition Research
Office of Research on Women's Health
Office of Science Policy
Office of AIDS Research

Full Proposal Deadline(s) (due by 5 p.m. submitting organization's local time):

October 03, 2025



Table Of Contents

[Summary of Program Requirements](#)

[I. Introduction](#)

[II. Program Description](#)

[III. Award Information](#)

[IV. Eligibility Information](#)

[V. Proposal Preparation and Submission Instructions](#)

[A. Proposal Preparation Instructions](#)

[B. Budgetary Information](#)

[C. Due Dates](#)

[D. Research.gov/Grants.gov Requirements](#)

[VI. NSF Proposal Processing and Review Procedures](#)

[A. Merit Review Principles and Criteria](#)

[B. Review and Selection Process](#)

[VII. Award Administration Information](#)

[A. Notification of the Award](#)

[B. Award Conditions](#)

C. Reporting Requirements

VIII. Agency Contacts

IX. Other Information

Important Information And Revision Notes

The Smart Health program solicitation has been revised and prospective Principal Investigators (PIs) are encouraged to read the solicitation carefully. Among the changes are the following:

- Themes areas have been revised;
- Changes have been made to the participating National Institutes of Health's Institutes and Centers; and,
- Proposal deadlines have been revised.

Any proposal submitted in response to this solicitation should be submitted in accordance with the *NSF Proposal & Award Policies & Procedures Guide* (PAPPG) that is in effect for the relevant due date to which the proposal is being submitted. The NSF PAPPG is regularly revised and it is the responsibility of the proposer to ensure that the proposal meets the requirements specified in this solicitation and the applicable version of the PAPPG. Submitting a proposal prior to a specified deadline does not negate this requirement.

Summary Of Program Requirements

General Information

Program Title:

Smart Health and Biomedical Research in the Era of Artificial Intelligence and Advanced Data Science (SCH)

Synopsis of Program:

The purpose of this interagency program solicitation is to support the development of transformative high-risk, high-reward advances in computer and information science, engineering, mathematics, statistics, behavioral and/or cognitive research to address pressing questions in the biomedical and public health communities. Transformations hinge on scientific and engineering innovations by interdisciplinary teams that develop novel methods to intuitively and intelligently collect, sense, connect, analyze and interpret data from individuals, devices and systems to enable discovery and optimize health. Solutions to these complex biomedical or public health problems demand the formation of interdisciplinary teams that are ready to address these issues, while advancing fundamental science and engineering.

Broadening Participation In STEM:

NSF has a mandate to broaden participation in science and engineering, as articulated and reaffirmed in law since 1950. Congress has charged NSF to "develop intellectual capital, both people and ideas, with particular emphasis on groups and regions that traditionally have not participated fully in science, mathematics, and engineering."

Cognizant Program Officer(s):

Please note that the following information is current at the time of publishing. See program website for any updates to the points of contact.

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- James Gao, National Eye Institute (NEI), NIH, telephone: 301-594-6074, email: james.gao@nih.gov
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- Noffisat Oki, National Institute of Dental and Craniofacial Research (NIDCR), telephone: 301-402-6778, email: noffisat.oki@nih.gov
- Susan Wright, National Institute on Drug Abuse (NIDA), NIH, telephone: 301-402-6683, email: Susan.wright@nih.gov
- Roger Miller, National Institute on Deafness and Other Communication Disorders (NIDCD), NIH, telephone: 301-402-3458, email: miller@nidcd.nih.gov
- Xujing Wang, National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), NIH, telephone: 301-451-2682, email: xujing.wang@nih.gov
- Christopher Duncan, National Institute of Environmental Health Sciences (NIEHS), NIH, telephone: 984-287-3256, email: christopher.duncan@NIH.gov
- Deborah Duran, National Institute on Minority Health and Health Disparities (NIMHD), NIH, telephone: 301-594-9809, email: deborah.duran@nih.gov
- Bill Duval, National Institute of Nursing Research (NINR), NIH, telephone: 301-435-0380, email: bill.duval@nih.gov
- Joseph Monaco, Brain Research Through Advancing Innovative Neurotechnologies (BRAIN) Initiative, telephone: 301-402-3823, email: Joseph.monaco@nih.gov
- Robert Cregg, Office of AIDS Research (OAR), telephone: 301-761-7557, email: Robert.Cregg@nih.gov
- David Leitman, National Institute of Mental Health (NIMH), telephone: 301-827-6131, email: david.leitman@nih.gov
- Jacqueline Lloyd, Office of Disease Prevention (ODP), telephone: 301-827-5559, email: lloydj2@nih.gov
- Nicholas Jury, Office of Nutrition Research (ONR), telephone: 301-827-1234, email: nicholas.jury@nih.gov

- Jamie White, Office of Research on Women's Health (ORWH), telephone: 301-496-9200, email: jamie.white@nih.gov
- Adam Berger, Office of Science Policy (OSP), telephone: 301-827-9676, email: adam.berger@nih.gov
- Christopher Hartshorn, National Center for Advancing Translational Sciences (NCATS), telephone: 301-402-0264, email: christopher.hartshorn@nih.gov
- Lyndon Joseph, National Institute on Aging (NIA), telephone: 301-496-6761, email: lyndon.joseph@nih.gov
- Goutham Reddy, National Library of Medicine (NLM), telephone: 301-827-6728, email: goutham.reddy@nih.gov
- Julia Berzhanskaya, National Heart, Lung, and Blood Institute (NHLBI), NIH, telephone: 301-443-3707, email: julia.berzhanskaya@nih.gov
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- Leslie C. Osborne, National Institute of Neurological Disorders and Stroke (NINDS), telephone: 240-921-135, email: leslie.osborne@nih.gov
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- Samantha Calabrese, Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD), NIH, telephone: 301-827-7568, email: samantha.calabrese@nih.gov

Applicable Catalog of Federal Domestic Assistance (CFDA) Number(s):

- 47.041 --- Engineering
- 47.049 --- Mathematical and Physical Sciences
- 47.070 --- Computer and Information Science and Engineering
- 47.075 --- Social Behavioral and Economic Sciences
- 93.121 --- National Institute of Dental and Craniofacial Research
- 93.173 --- National Institute on Deafness and Other Communication Disorders
- 93.213 --- National Center for Complementary and Integrative Health
- 93.242 --- National Institute of Mental Health
- 93.279 --- National Institute on Drug Abuse
- 93.286 --- National Institute of Biomedical Imaging and Bioengineering
- 93.310 --- NIH Office of Data Science Strategy
- 93.350 --- National Center for Advancing Translational Sciences
- 93.361 --- National Institute of Nursing Research
- 93.396 --- National Cancer Institute
- 93.846 --- National Institute of Arthritis and Musculoskeletal and Skin Disease
- 93.847 --- National Institute of Diabetes and Digestive and Kidney Diseases
- 93.853 --- National Institute of Neurological Disorders and Stroke
- 93.865 --- Eunice Kennedy Shriver National Institute of Child Health and Human Development
- 93.866 --- National Institute on Aging
- 93.867 --- National Eye Institute

- 93.879 --- National Library of Medicine

Award Information

Anticipated Type of Award: Standard Grant or Continuing Grant or Cooperative Agreement

Estimated Number of Awards: 10 to 16 per year, subject to the availability of funds.

Projects will be funded for up to four years for a total of \$1,200,000 (\$300,000 per year).

Anticipated Funding Amount: \$15,000,000 to \$20,000,000 will be invested in proposals submitted to this solicitation in each year of the solicitation, subject to the availability of funds and the quality of the proposals received.

Eligibility Information**Who May Submit Proposals:**

Proposals may only be submitted by the following:

- Institutions of Higher Education (IHEs): Two- and four-year IHEs (including community colleges) accredited in, and having a campus located in the US, acting on behalf of their faculty members. Special Instructions for International Branch Campuses of US IHEs: If the proposal includes funding to be provided to an international branch campus of a US institution of higher education (including through use of sub-awards and consultant arrangements), the proposer must explain the benefit(s) to the project of performance at the international branch campus, and justify why the project activities cannot be performed at the US campus.
- Non-profit, non-academic organizations: Independent museums, observatories, research laboratories, professional societies and similar organizations located in the U.S. that are directly associated with educational or research activities.

Who May Serve as PI:

There are no restrictions or limits.

Limit on Number of Proposals per Organization:

There are no restrictions or limits.

Limit on Number of Proposals per PI or co-PI: 2

In each annual competition, an investigator may participate as Principal Investigator (PI), co-Principal Investigator (co-PI), Project Director (PD), Senior/Key Personnel or Consultant in **no more than two** proposals submitted in response to each annual deadline in this solicitation. **These eligibility constraints will be strictly enforced in order to treat everyone fairly and consistently.** In the event that an individual exceeds this limit, proposals received within the limit will be accepted based on earliest date and time of proposal submission (i.e., the first two proposals received will be accepted, and the remainder will be returned without review). **No exceptions will be made.**

Proposals submitted in response to this solicitation may not duplicate or be substantially similar to other proposals concurrently under consideration by NSF or NIH programs or study sections. Duplicate or substantially similar proposals will be returned without review. NIH will not accept any application that is essentially the same as one already reviewed within the past 37 months (as described in the [NIH Grants Policy Statement](#)), except for submission:

- To an NIH Requests for Applications (RFA) of an application that was submitted previously as an investigator-initiated application but not paid;
- Of an NIH investigator-initiated application that was originally submitted to an RFA but not paid; or

- Of an NIH application with a changed grant activity code.

Proposal Preparation and Submission Instructions

A. Proposal Preparation Instructions

- **Letters of Intent:** Not required
- **Preliminary Proposal Submission:** Not required
- **Full Proposals:**
 - Full Proposals submitted via Research.gov: *NSF Proposal and Award Policies and Procedures Guide (PAPPG)* guidelines apply. The complete text of the PAPPG is available electronically on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.
 - Full Proposals submitted via Grants.gov: *NSF Grants.gov Application Guide: A Guide for the Preparation and Submission of NSF Applications via Grants.gov* guidelines apply (Note: The *NSF Grants.gov Application Guide* is available on the Grants.gov website and on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=grantsgovguide).

B. Budgetary Information

- **Cost Sharing Requirements:**

Inclusion of voluntary committed cost sharing is prohibited.
- **Indirect Cost (F&A) Limitations:**

For NSF, Proposal & Award Policies & Procedures Guide (PAPPG) Guidelines apply.

For NIH, indirect costs on foreign subawards/subcontracts will be limited to eight (8) percent.
- **Other Budgetary Limitations:**

Other budgetary limitations apply. Please see the full text of this solicitation for further information.

C. Due Dates

- **Full Proposal Deadline(s)** (due by 5 p.m. submitting organization's local time):

October 03, 2025

Proposal Review Information Criteria

Merit Review Criteria:

National Science Board approved criteria apply.

Award Administration Information

Award Conditions:

Additional award conditions apply. Please see the full text of this solicitation for further information.

Reporting Requirements:

Additional reporting requirements apply. Please see the full text of this solicitation for further information.

I. Introduction

The need for a significant transformation in medical, public health and healthcare delivery approaches has been recognized by numerous organizations and captured in a number of reports. For example, the Networking and Information Technology Research and Development (NITRD) program released the [Federal Health Information Technology Research and Development Strategic Framework](#) in 2020 that pointed to an overwhelming need for the integration between the computing, informatics, engineering, mathematics and statistics, behavioral and social science disciplines, and the biomedical, and public health research communities to produce the innovation necessary to improve the health of the country. Recent developments and significant advances in machine learning (ML), artificial intelligence (AI), deep learning, high performance and cloud computing, and availability of new datasets make such integration achievable, as documented in the 2023 updated [National Artificial Intelligence Research and Development Strategic Plan](#).

These anticipated transformations hinge on scientific and engineering innovations by interdisciplinary teams that intelligently collect, connect, analyze and interpret data from individuals, devices, and systems to enable discovery and optimize health. Technical challenges include a range of issues, including effective data generation, analysis and automation for a range of biomedical devices (from imaging through medical devices) and systems (e.g., electronic health records) and consumer devices (including the Internet of Things), as well as new technology to generate knowledge. Underlying these challenges are many fundamental scientific and engineering issues that require investment in interdisciplinary research to actualize the transformations, which is the goal of this solicitation.

II. Program Description

This interagency solicitation is a collaboration between NSF and the NIH. The Smart Health program supports innovative, high-risk/high-reward research with the promise of disruptive transformations in biomedical and public health research, which can only be achieved by well-coordinated, convergent, and interdisciplinary approaches that draw from multiple domains of computer and information science, engineering, mathematical sciences and the biomedical, social, behavioral, and economic sciences. Therefore, the work to be funded by this solicitation **must make fundamental scientific or engineering contributions to two or more disciplines**, such as computer or information sciences, engineering, mathematical sciences, statistics, social, behavioral, or cognitive sciences to improve fundamental understanding of human biological, biomedical, public health and/or health-related processes and address a key health problem. The research teams must include members with appropriate and demonstrable expertise in the major areas involved in the work. Traditional disease-centric medical, clinical, pharmacological, biological or physiological studies and evaluations are outside the scope of this solicitation. In addition, fundamental biological research with humans that also does not advance other fundamental science or engineering areas is out of scope for this program. Finally, proposals addressing health indirectly in the education or work environment are also out of scope.

Generating these transformations will require fundamental research and development of new tools, workflows and methods across many dimensions; some of the themes are highlighted below. These themes should be seen as examples and not exhaustive.

1. Fairness and Trustworthiness: Advancing fairness and trustworthiness in modeling in AI/ML is a highly interdisciplinary endeavor. Real world considerations go beyond the analytics and can inform new directions for computational science to better realize the benefits of algorithmic and data fairness and trustworthiness. The complexity of biomedical and health systems requires deeper understanding of causality in AI/ML models; new ways of integrating social and economic data to optimize health, such as disease heterogeneity, disease prevention, resilience, and treatment response, while systematically accounting for a broad range of uncertainties; and new insights into human-AI systems for clinical decision support. In general, this thrust supports the conduct of fundamental computational research into theories, techniques, and methodologies that go well beyond today's capabilities and are motivated by challenges and requirements in biomedical applications.

2. Transformative Analytics in Biomedical and Behavioral Research: As biomedical and behavioral research continues to generate large amount of multi-level and multi-scale data (e.g., clinical, imaging, personal, social, contextual, environmental, and organizational data), challenges remain. New development in areas such as artificial intelligence and machine learning (AI/ML), natural language technologies (NLT), mathematics and statistics and/or quantum information science (QIS) also bring opportunities to address important biomedical and behavioral problems. This theme will support

efforts to push forward the current frontline of AI/ML and advanced analytics for biomedical and behavioral research including:

- novel data reduction methods;
- new robust knowledge representations, visualizations, reasoning algorithms, optimization, modeling and inference methods to support development of innovative models for the study of health and disease;
- new computational approaches with provable mathematical guarantees for fusion and analysis of complex behavioral, biomedical and image data to improve inference accuracy, especially in scenarios of noisy and limited data records;
- novel explainable and interpretable AI/ML model development;
- advanced data management systems with the capability to deal with security, privacy and provenance issues;
- novel data systems to build a connected and modernized biomedical data ecosystem;
- development of novel technologies to extract information from unstructured text data such as clinical notes, radiology and pathology reports;
- development of novel simulation and modeling methods to aid in the design and evaluation of new assessments, treatments and medical devices; and
- novel QIS approaches to unique challenges in biomedical and behavioral research.

3. Next Generation Multimodal and Reconfigurable Sensing Systems: This theme addresses the need for new multimodal and reconfigurable sensing systems/platforms and analytics to generate predictive and personalized models of health. The next generation of sensor systems for smart health must have just-in-time monitoring of biomarkers from multiple sensor modalities (e.g., electrochemical, electromagnetic, mechanical, optical, acoustic, etc.) interfaced with different platforms (e.g., mobile, wearable, and implantable). Existing sensor systems generally operate either in discrete formats or with limited inter-connectivity, and are limited in accuracy, selectivity, reliability and data throughput. Those limitations can be overcome by integrating heterogeneous sensing modalities and by having field-adaptive reconfigurable sensor microsystems. This theme encourages the design and fabrication of multimodal and/or reconfigurable sensor systems through innovative research on novel functional materials, devices and circuits for sensing or active interrogation of system states, imaging, communications, and computing. Areas of interest include miniaturized sensor microsystems with integrated signal processing and communication functionalities. Another area of interest is multimodal or reconfigurable sensor systems with dramatically reduced power consumption to extend battery lifetime and enable self-powered operation, making the sensor systems suitable for wearable and implantable applications. Other areas of interest include real-time monitoring of analytes and new biorecognition elements that can be reconfigured to target different analytes on-demand. This thrust also requires researchers to integrate data generated by the multimodal sensor systems, as well as data from other sources, such as laboratory generated data (e.g., genomics, proteomics, etc.), patient-reported outcomes, electronic health records, and existing data sources.

4. Cyber-Physical Systems: Development and adoption of automation has lagged in the biomedical and public health communities. Cyber-physical systems (CPS) are controlled systems built from, and dependent upon, the seamless integration of computation and physical components. These data-rich systems enable new and higher degrees of automation and autonomy. Thus, this theme supports work that enables the creation of closed-loop or human-in-the-loop CPS systems to assess, treat and reduce adverse health events or behaviors, with core research areas including control, data analytics, and machine learning including real-time learning for control, autonomy, design, Internet of Things (IoT), networking, privacy, real-time systems, safety, security, and verification. Finally, development of automated technology that can be utilized across a range of settings (e.g., home, primary care, schools, criminal justice system, child welfare agencies, community-based organizations) to optimize the delivery of effective health interventions is also within scope of the theme.

5. Robotics: This theme addresses the need for novel robotics to provide support and/or automation to enhance health, lengthen lifespan and reduce illness, enhance social connectedness and reduce disabilities. The theme encourages research on robotic systems that exhibit significant levels of both computational capability and physical complexity. Robots are defined as intelligence embodied in an engineered construct, with the ability to process information, sense,

plan, and move within or substantially alter its working environment. Here intelligence includes a broad class of methods that enable a robot to solve problems or to make contextually appropriate decisions and act upon them. Currently, robotic devices in health have focused on limited areas (e.g., surgical robotics and exoskeletons). This theme welcomes a wide range of robotic areas, as well as research that considers inextricably interwoven questions of intelligence, computation, and embodiment. The next generation of robotic systems for smart health will need to also have to consider human-robot interaction to enhance usability and effectiveness.

6. Biomedical Image interpretation. This theme's goal is to determine how characteristics of human pattern recognition, visual search, perceptual learning, attentional biases, etc. can inform and improve image interpretation. This theme would include using and developing presentation modalities (e.g., pathologists reading optical slides through a microscope vs. digital whole-slide imagery) and identifying the sources of inter- and intra-observer variability. The theme encourages development of models of how multi-modal contextual information (e.g., integrating patient history, omics, etc. with imaging data) changes the perception of complex images. It also supports new methods to exploit experts' implicit knowledge to improve perceptual decision making (e.g., via rapid gist extraction, context-guided search, etc.). Research on optimal methods for conveying 3D (and 4D) information about anatomy and physiology to human observers is also welcome. This theme also supports advances in image data compression algorithm development to enable more efficient data storage.

The above listed themes are to provide examples for possible research activities that may be supported by this solicitation, but by no means should the proposed research activities be restricted to these themes. These research themes are clearly not mutually exclusive, and a given project may address multiple themes. This solicitation aims to support research activities that complement rather than duplicate the core programs of the NSF Directorates and the NIH Institutes and Centers and the research efforts supported by other agencies such as the Agency for Healthcare Research and Quality.

NSF supports investigation of fundamental research questions with broadly applicable results. The Smart Health program supports research evaluation with humans. Because advancing fundamental science is early-stage research, [randomized control trials](#) are not appropriate for this solicitation and will not be funded. Research that has advanced to a stage that requires randomized control trials should be submitted to an agency whose mission is to improve health.

NIH supports research and discovery that improve human health and save lives. This joint program focuses on fundamental research of generalizable, disease-agnostic approaches with broadly applicable results that align with NIH's [Strategic Plan for Data Science](#).

Integrative Innovation:

Proposals submitted to this solicitation must be integrative and undertake research addressing key application areas by solving problems in multiple scientific domains. The work must make **fundamental scientific or engineering contributions to two or more disciplines**, such as computer or information sciences, engineering, mathematical sciences, social, behavioral, cognitive and/or economic sciences and address a key health problem. For example, these projects are expected to advance understanding of how computing, engineering and mathematics, combined with advances in behavioral and social science research, would support transformations in health, medicine and/or healthcare and improve the quality of life. Projects are expected to include students and postdocs. Project descriptions must be comprehensive and well-integrated, and should make a convincing case that the collaborative contributions of the project team will be greater than the sum of each of their individual contributions. Collaborations with researchers in the health application domains are required. Such collaborations typically involve multiple institutions, but this is not required. Because the successes of collaborative research efforts are known to depend on thoughtful collaboration mechanisms that regularly bring together the various participants of the project, **a Collaboration Plan is required for ALL proposals**. Projects will be funded for up to a four-year period and for up to a total of \$300,000 per year. The proposed budget should be commensurate with the corresponding scope of work. Rationale must be provided to explain why a budget of the requested size is required to carry out the proposed work.

III. Award Information

Estimated program budget, number of awards and average award size/duration are subject to the availability of funds. An estimated 10 to 16 projects will be funded, subject to availability of funds. Up to \$15,000,000-20,000,000 of NSF funds will be invested in proposals submitted to this solicitation. The number of awards and program budgets are subject to the availability of funds.

All awards under this solicitation made by NSF will be as grants or cooperative agreements as determined by the supporting agency. All awards under this solicitation made by NIH will be as grants or cooperative agreements.

Scientists from all disciplines are encouraged to participate. Projects will be awarded depending on the availability of funds and with consideration for creating a balanced overall portfolio.

IV. Eligibility Information

Who May Submit Proposals:

Proposals may only be submitted by the following:

- Institutions of Higher Education (IHEs): Two- and four-year IHEs (including community colleges) accredited in, and having a campus located in the US, acting on behalf of their faculty members. Special Instructions for International Branch Campuses of US IHEs: If the proposal includes funding to be provided to an international branch campus of a US institution of higher education (including through use of sub-awards and consultant arrangements), the proposer must explain the benefit(s) to the project of performance at the international branch campus, and justify why the project activities cannot be performed at the US campus.
- Non-profit, non-academic organizations: Independent museums, observatories, research laboratories, professional societies and similar organizations located in the U.S. that are directly associated with educational or research activities.

Who May Serve as PI:

There are no restrictions or limits.

Limit on Number of Proposals per Organization:

There are no restrictions or limits.

Limit on Number of Proposals per PI or co-PI: 2

In each annual competition, an investigator may participate as Principal Investigator (PI), co-Principal Investigator (co-PI), Project Director (PD), Senior/Key Personnel or Consultant in **no more than two** proposals submitted in response to each annual deadline in this solicitation. **These eligibility constraints will be strictly enforced in order to treat everyone fairly and consistently.** In the event that an individual exceeds this limit, proposals received within the limit will be accepted based on earliest date and time of proposal submission (i.e., the first two proposals received will be accepted, and the remainder will be returned without review). **No exceptions will be made.**

Proposals submitted in response to this solicitation may not duplicate or be substantially similar to other proposals concurrently under consideration by NSF or NIH programs or study sections. Duplicate or substantially similar proposals will be returned without review. NIH will not accept any application that is essentially the same as one already reviewed within the past 37 months (as described in the [NIH Grants Policy Statement](#)), except for submission:

- To an NIH Requests for Applications (RFA) of an application that was submitted previously as an investigator-initiated application but not paid;
- Of an NIH investigator-initiated application that was originally submitted to an RFA but not paid; or

- Of an NIH application with a changed grant activity code.

V. Proposal Preparation And Submission Instructions

A. Proposal Preparation Instructions

Full Proposal Preparation Instructions: Proposers may opt to submit proposals in response to this Program Solicitation via Research.gov or Grants.gov.

- Full Proposals submitted via Research.gov: Proposals submitted in response to this program solicitation should be prepared and submitted in accordance with the general guidelines contained in the *NSF Proposal and Award Policies and Procedures Guide* (PAPPG). The complete text of the PAPPG is available electronically on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg. Paper copies of the PAPPG may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov. The Prepare New Proposal setup will prompt you for the program solicitation number.
- Full proposals submitted via Grants.gov: Proposals submitted in response to this program solicitation via Grants.gov should be prepared and submitted in accordance with the *NSF Grants.gov Application Guide: A Guide for the Preparation and Submission of NSF Applications via Grants.gov*. The complete text of the *NSF Grants.gov Application Guide* is available on the Grants.gov website and on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=grantsgovguide. To obtain copies of the Application Guide and Application Forms Package, click on the Apply tab on the Grants.gov site, then click on the Apply Step 1: Download a Grant Application Package and Application Instructions link and enter the funding opportunity number, (the program solicitation number without the NSF prefix) and press the Download Package button. Paper copies of the Grants.gov Application Guide also may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov.

In determining which method to utilize in the electronic preparation and submission of the proposal, please note the following:

Collaborative Proposals. All collaborative proposals submitted as separate submissions from multiple organizations must be submitted via Research.gov. PAPPG Chapter II.E.3 provides additional information on collaborative proposals.

See PAPPG Chapter II.D.2 for guidance on the required sections of a full research proposal submitted to NSF. Please note that the proposal preparation instructions provided in this program solicitation may deviate from the PAPPG instructions.

The following information SUPPLEMENTS (note that it does NOT replace) the guidelines provided in the NSF Proposal & Award Policies & Procedures Guide (PAPPG).

Proposal Titles: Proposal titles must begin with **SCH**, followed by a colon and the title of the project (i.e., **SCH: Title**). If you submit a proposal as part of a set of collaborative proposals, the title of the proposal should begin with **Collaborative Research** followed by a colon, then **SCH** followed by a colon, and the title. For example, if you are submitting a collaborative set of proposals, then the title of each would be **Collaborative Research: SCH: Title**.

Proposals from PIs in institutions that have Research in Undergraduate Institutions (RUI) eligibility should have a proposal title that begins with **Collaborative Research** (if applicable), followed by a colon, then **SCH** followed by a colon, then **RUI** followed by a colon, and then the title, for example, **Collaborative Research: SCH: RUI: Title**.

Project Summary (1 page limit): The Project Summary must include three labeled sections: Overview, Intellectual Merit and Broader Impacts. At the beginning of the Overview section of the Project Summary enter the title of the Smart Health project, the name of the PI and the lead institution. The overview includes a description of the project. Intellectual Merit should describe the transformative research and the potential of the proposed activity to advance knowledge. Broader Impacts should describe the potential of the proposed activity to benefit society and contribute to the achievement of specific, desired societal outcomes. The Broader Impacts can include education goals, and the community (communities) that will be impacted by its results.

Project Description: There is a 15-page limit for all proposals. Within the project description, **include a section labeled 'Evaluation Plan' that includes a description of how the team will evaluate the proposed science/engineering.** This plan could include results from applications of the research to specific outcomes in health domain, efficacy studies, assessments of learning and engagement, and other such evaluation. The proposed **Evaluation Plan** should be appropriate for the size and scope of the project.

Please note that the Collaboration Plan must be submitted as a Supplementary Document for this solicitation; see guidance below.

Proposal Budget: It is expected that the PIs, co-PIs, and other team members funded by the project will attend an SCH PI meeting annually to present project research findings and capacity-building or community outreach activities. Requested budgets should include funds for travel to this annual event for at least one project PI.

Supplementary Documents: In the Supplementary Documents Section, upload the following:

1. **Collaboration Plan.** Proposals **must** include a Collaboration Plan. **The Collaboration Plan must be submitted as a supplementary document and cannot exceed two pages. Proposals that do not include a Collaboration Plan will be returned without review.** The Collaboration Plan must be labeled "Collaboration Plan" and must include: 1) the specific roles of the collaborating PIs, co-PIs, other Senior/Key Personnel and paid consultants at all organizations involved; 2) how the project will be managed across institutions and disciplines; 3) identification of the specific collaboration mechanisms that will enable cross-institution and/or cross-discipline scientific integration (e.g., workshops, graduate student exchange, project meetings at conferences, use of videoconferencing and other communication tools, software repositories, etc.); and 4) specific references to the budget line items that support these collaboration mechanisms.
2. **Human Subjects Protection.** Proposals involving human subjects should include a supplementary document of no more than two pages in length summarizing potential risks to human subjects; plans for recruitment and informed consent; inclusion of women, minorities, and children; and planned procedures to protect against or minimize potential risks. **Human subjects plans must include the NIH enrollment table** (https://era.nih.gov/erahelp/assist/Content/ASSIST_Help_Topics/3_Form_Screens/PHS_HS_CT/Incl_Enroll_Rprt.htm, please see the Planned enrollment table for the expected format).
3. **Vertebrate Animals.** Proposals involving vertebrate animals should include a supplementary document of no more than two pages in length that addresses the following points:
 - a. Detailed description and justification of the proposed use of the animals, including species, strains, ages, sex, and number to be used;
 - b. Information on the veterinary care of the animals;
 - c. Description of procedures for minimizing discomfort, distress, pain, and injury; and
 - d. Method of euthanasia and the reasons for its selection.
4. **Data Management and Sharing Plan (required).** This supplementary document should describe how the proposal will conform to NSF policy on the dissemination and sharing of research results.

See Chapter II.D.2 of the PAPPG for full policy implementation and the section on Data Management and Sharing Plans.

For additional information on the Dissemination and Sharing of Research Results, see:

<https://www.nsf.gov/bfa/dias/policy/dmp.jsp>.

For specific guidance for Data Management and Sharing Plans submitted to the Directorate for Computer and Information Science and Engineering (CISE) see: https://www.nsf.gov/cise/cise_dmp.jsp.

5. **Documentation of Collaborative Arrangements of Significance to the Proposal through Letters of Collaboration.** There are two types of collaboration, one involving individuals/organizations that are included in the budget, and the other involving individuals/organizations that are not included in the budget. Collaborations that are included in the budget should be described in the Project Description and the Collaboration Plan.

Any substantial collaboration with individuals/organizations not included in the budget should be described in the Facilities, Equipment and Other Resources section of the proposal (see PAPPG Chapter II.D.2.g). In either case, whether or not the collaborator is included in the budget, **a letter of collaboration from each named participating organization other than the submitting lead, non-lead, and/or subawardee institutions must be provided at the time of submission of the proposal. Such letters describe the nature of the collaboration and what the collaborator(s) brings to the project. They must explicitly state the nature of the collaboration, appear on the organization's letterhead, and be signed by the appropriate organizational representative. These letters must not otherwise deviate from the restrictions and requirements set forth in the PAPPG, Chapter II.D.2.i.**

Please note that letters of support, that do not document collaborative arrangements of significance to the project, but primarily convey a sense of enthusiasm for the project and/or highlight the qualifications of the PI or co-PI may not be submitted, and reviewers will be instructed not to consider these letters in reviewing the merits of the proposal.

6. **List of Project Personnel and Partner Institutions (Note - In collaborative proposals, only the lead institution should provide this information).** Provide current, accurate information for all personnel and institutions involved in the project. NSF staff will use this information in the merit review process to manage reviewer selection. The list should include all PIs, co-PIs, Senior/Key Personnel, paid/unpaid Consultants or Collaborators, Subawardees, Postdocs, and project-level advisory committee members. This list should be numbered and include (in this order) Full name, Organization(s), and Role in the project, with each item separated by a semi-colon. Each person listed should start a new numbered line. For example:

1. Mei Lin; XYZ University; PI
2. Jak Jabes; University of PQR; Senior/Key Personnel
3. Jane Brown; XYZ University; Postdoctoral Researcher
4. Rakel Ademas; ABC Inc.; Paid Consultant
5. Maria Wan; Welldone Institution; Unpaid Collaborator
6. Rimon Greene; ZZZ University; Subawardee

7. **Mentoring Plan (if applicable).** Each proposal that requests funding to support postdoctoral scholars or graduate students must include, as a supplementary document, a description of the mentoring activities that will be provided for such individuals. Please be advised that if required, NSF systems will not permit submission of a proposal that is missing a Mentoring Plan. See Chapter II.D.2.i of the PAPPG for further information about the implementation of this requirement.

8. **Other Specialized Information.** RUI Proposals: PIs from predominantly undergraduate institutions should follow the instructions <https://www.nsf.gov/funding/opportunities/rui-roa-pui-facilitating-research-predominantly-undergraduate>.

Single Copy Documents:

(1) Collaborators and Other Affiliations Information.

Proposers should follow the guidance specified in Chapter II.D.2.h of the NSF PAPPG. Grants.gov Users: The COA information must be provided through use of the COA template and uploaded as a PDF attachment.

Note the distinction to the list of Project Personnel and Partner Institutions specified above under Supplementary Documents: the listing of all project participants is collected by the project lead and entered as a Supplementary Document, which is then automatically included with all proposals in a project. The Collaborators and Other Affiliations are entered for each participant within each proposal and, as Single Copy Documents, are available only to NSF staff.

SCH Proposal Preparation Checklist:

The following checklist is provided as a reminder of the items that should be verified before submitting a proposal to this solicitation. This checklist is a summary of the requirements described above and in the PAPPG and **does not replace the complete set of requirements in the PAPPG**. For the items marked with (RWR), the proposal will be returned without review if the required item is noncompliant at the submission deadline.

- (RWR) A two-page Collaboration Plan must be included as a Supplementary Document.
- Letters of Collaboration are permitted as Supplementary Documents.
- (RWR) Within the Project Description, a section labeled "Broader Impacts" that describes the potential to benefit society and contribute to the achievement of specific, desired societal outcomes.
- Within the Project Description, a section labeled Evaluation Plan that details how the project will be evaluated.
- (RWR) Within the Project Description, a description of "Results from Prior NSF Support".
- A list of Project Personnel and Partner Institutions is required as a Supplementary Document.
- Proposals involving human subjects or animals, should include a Human Subjects Plan with an NIH Enrollment Table or Vertebrate Animals Plan as a Supplementary Document.
- Collaborators & Other Affiliations (COA) for each PI, co-PI, and Senior/Key Personnel should be submitted using the spreadsheet template uploaded as Single Copy Documents.

B. Budgetary Information

Cost Sharing:

Inclusion of voluntary committed cost sharing is prohibited.

Indirect Cost (F&A) Limitations:

For NSF, Proposal & Award Policies & Procedures Guide (PAPPG) Guidelines apply.

For NIH, indirect costs on foreign subawards/subcontracts will be limited to eight (8) percent.

Other Budgetary Limitations:

Budgets should include travel funds to attend one SCH PI meeting annually for the project PIs, co-PIs and other team members as appropriate from all collaborating institutions. Requested budgets should include funds for travel to this annual event for at least one project PI.

C. Due Dates

- **Full Proposal Deadline(s)** (due by 5 p.m. submitting organization's local time):

October 03, 2025

D. Research.gov/Grants.gov Requirements

For Proposals Submitted Via Research.gov:

To prepare and submit a proposal via Research.gov, see detailed technical instructions available at: https://www.research.gov/research-portal/appmanager/base/desktop?_nfpb=true&_pageLabel=research_node_display&_nodePath=/researchGov/Service/Desktop/ProposalPreparationandSubmission.html. For Research.gov user support, call the Research.gov Help Desk at 1-800-381-1532 or e-mail rgov@nsf.gov. The Research.gov Help Desk answers general technical questions related to the use of the Research.gov system. Specific questions related to this program solicitation should be referred to the NSF program staff contact(s) listed in Section VIII of this funding opportunity.

For Proposals Submitted Via Grants.gov:

Before using Grants.gov for the first time, each organization must register to create an institutional profile. Once registered, the applicant's organization can then apply for any federal grant on the Grants.gov website. Comprehensive information about using Grants.gov is available on the Grants.gov Applicant Resources webpage: <https://www.grants.gov/applicants>. In addition, the NSF Grants.gov Application Guide (see link in Section V.A) provides instructions regarding the technical preparation of proposals via Grants.gov. For Grants.gov user support, contact the Grants.gov Contact Center at 1-800-518-4726 or by email: support@grants.gov. The Grants.gov Contact Center answers general technical questions related to the use of Grants.gov. Specific questions related to this program solicitation should be referred to the NSF program staff contact(s) listed in Section VIII of this solicitation.

Submitting the Proposal: Once all documents have been completed, the Authorized Organizational Representative (AOR) must submit the application to Grants.gov and verify the desired funding opportunity and agency to which the application is submitted. The AOR must then sign and submit the application to Grants.gov. The completed application will be transferred to Research.gov for further processing.

The NSF [Grants.gov Proposal Processing in Research.gov informational page](#) provides submission guidance to applicants and links to helpful resources including the NSF [Grants.gov Application Guide](#), [Grants.gov Proposal Processing in Research.gov how-to guide](#), and [Grants.gov Submitted Proposals Frequently Asked Questions](#). Grants.gov proposals must pass all NSF pre-check and post-check validations in order to be accepted by Research.gov at NSF.

When submitting via Grants.gov, NSF strongly recommends applicants initiate proposal submission at least five business days in advance of a deadline to allow adequate time to address NSF compliance errors and resubmissions by 5:00 p.m. submitting organization's local time on the deadline. Please note that some errors cannot be corrected in Grants.gov. Once a proposal passes pre-checks but fails any post-check, an applicant can only correct and submit the in-progress proposal in Research.gov.

Proposers that submitted via Research.gov may use Research.gov to verify the status of their submission to NSF. For proposers that submitted via Grants.gov, until an application has been received and validated by NSF, the Authorized Organizational Representative may check the status of an application on Grants.gov. After proposers have received an e-mail notification from NSF, Research.gov should be used to check the status of an application.

VI. NSF Proposal Processing And Review Procedures

Proposals received by NSF are assigned to the appropriate NSF program for acknowledgement and, if they meet NSF requirements, for review. All proposals are carefully reviewed by a scientist, engineer, or educator serving as an NSF Program Officer, and usually by three to ten other persons outside NSF either as *ad hoc* reviewers, panelists, or both, who are experts in the particular fields represented by the proposal. These reviewers are selected by Program Officers charged with oversight of the review process. Proposers are invited to suggest names of persons they believe are especially well qualified to review the proposal and/or persons they would prefer not review the proposal. These suggestions may serve as one source in the reviewer selection process at the Program Officer's discretion. Submission of such names, however, is optional. Care is taken to ensure that reviewers have no conflicts of interest with the proposal. In addition, Program Officers may obtain comments from site visits before recommending final action on proposals. Senior NSF staff further review recommendations for awards. A flowchart that depicts the entire NSF proposal and award process (and associated timeline) is included in PAPPG Exhibit III-1.

A comprehensive description of the Foundation's merit review process is available on the NSF website at: <https://www.nsf.gov/funding/overview>.

A. Merit Review Principles and Criteria

The National Science Foundation strives to invest in a robust and diverse portfolio of projects that creates new knowledge and enables breakthroughs in understanding across all areas of science and engineering research and education. To

identify which projects to support, NSF relies on a merit review process that incorporates consideration of both the technical aspects of a proposed project and its potential to contribute more broadly to advancing NSF's mission "to promote the progress of science; to advance the national health, prosperity, and welfare; to secure the national defense; and for other purposes." NSF makes every effort to conduct a fair, competitive, transparent merit review process for the selection of projects.

1. Merit Review Principles

These principles are to be given due diligence by PIs and organizations when preparing proposals and managing projects, by reviewers when reading and evaluating proposals, and by NSF program staff when determining whether or not to recommend proposals for funding and while overseeing awards. Given that NSF is the primary federal agency charged with nurturing and supporting excellence in basic research and education, the following three principles apply:

- All NSF projects should be of the highest quality and have the potential to advance, if not transform, the frontiers of knowledge.
- NSF projects, in the aggregate, should contribute more broadly to achieving societal goals. These "Broader Impacts" may be accomplished through the research itself, through activities that are directly related to specific research projects, or through activities that are supported by, but are complementary to, the project. The project activities may be based on previously established and/or innovative methods and approaches, but in either case must be well justified.
- Meaningful assessment and evaluation of NSF funded projects should be based on appropriate metrics, keeping in mind the likely correlation between the effect of broader impacts and the resources provided to implement projects. If the size of the activity is limited, evaluation of that activity in isolation is not likely to be meaningful. Thus, assessing the effectiveness of these activities may best be done at a higher, more aggregated, level than the individual project.

With respect to the third principle, even if assessment of Broader Impacts outcomes for particular projects is done at an aggregated level, PIs are expected to be accountable for carrying out the activities described in the funded project. Thus, individual projects should include clearly stated goals, specific descriptions of the activities that the PI intends to do, and a plan in place to document the outputs of those activities.

These three merit review principles provide the basis for the merit review criteria, as well as a context within which the users of the criteria can better understand their intent.

2. Merit Review Criteria

All NSF proposals are evaluated through use of the two National Science Board approved merit review criteria. In some instances, however, NSF will employ additional criteria as required to highlight the specific objectives of certain programs and activities.

The two merit review criteria are listed below. **Both** criteria are to be given **full consideration** during the review and decision-making processes; each criterion is necessary but neither, by itself, is sufficient. Therefore, proposers must fully address both criteria. (PAPPG Chapter II.D.2.d(i). contains additional information for use by proposers in development of the Project Description section of the proposal). Reviewers are strongly encouraged to review the criteria, including PAPPG Chapter II.D.2.d(i), prior to the review of a proposal.

When evaluating NSF proposals, reviewers will be asked to consider what the proposers want to do, why they want to do it, how they plan to do it, how they will know if they succeed, and what benefits could accrue if the project is successful. These issues apply both to the technical aspects of the proposal and the way in which the project may make broader contributions. To that end, reviewers will be asked to evaluate all proposals against two criteria:

- **Intellectual Merit:** The Intellectual Merit criterion encompasses the potential to advance knowledge; and
- **Broader Impacts:** The Broader Impacts criterion encompasses the potential to benefit society and contribute to the achievement of specific, desired societal outcomes.

The following elements should be considered in the review for both criteria:

1. What is the potential for the proposed activity to
 - a. Advance knowledge and understanding within its own field or across different fields (Intellectual Merit); and
 - b. Benefit society or advance desired societal outcomes (Broader Impacts)?
2. To what extent do the proposed activities suggest and explore creative, original, or potentially transformative concepts?
3. Is the plan for carrying out the proposed activities well-reasoned, well-organized, and based on a sound rationale? Does the plan incorporate a mechanism to assess success?
4. How well qualified is the individual, team, or organization to conduct the proposed activities?
5. Are there adequate resources available to the PI (either at the home organization or through collaborations) to carry out the proposed activities?

Broader impacts may be accomplished through the research itself, through the activities that are directly related to specific research projects, or through activities that are supported by, but are complementary to, the project. NSF values the advancement of scientific knowledge and activities that contribute to achievement of societally relevant outcomes. Such outcomes include, but are not limited to improved STEM education and educator development at any level; increased public scientific literacy and public engagement with science and technology; improved well-being of individuals in society; development of a globally competitive STEM workforce; increased partnerships between academia, industry, and others; improved national security; increased economic competitiveness of the United States; and enhanced infrastructure for research and education.

Proposers are reminded that reviewers will also be asked to review the Data Management and Sharing Plan and the Mentoring Plan, as appropriate.

B. Review and Selection Process

Proposals submitted in response to this program solicitation will be reviewed by Ad hoc Review and/or Panel Review, or Joint agency review.

Review Process and Deviations from the NSF PAPPG

This section provides agency-specific guidance for the SCH program.

NSF will take the lead in organizing and conducting the review process in compliance with the Federal Advisory Committee Act.

In addition to any conflict forms required by NSF, an NIH Post-Review Certification Form will be circulated at or near the end of the second day of the review meeting and collected by the NIH Scientific Review Officer (SRO). By signing the Post-Review Certification Form, panelists will certify for NIH that confidentiality and conflict-of-interest procedures have been followed. Conflicts of interest are handled in a manner similar to NSF procedures: those in conflict will be asked to step out of the room, or as appropriate, NSF's Designated Ethics Official or designee may recommend remedies to resolve specific conflicts on a case by case basis. Co-investigators and investigators that would directly benefit should the grant be awarded are ineligible to serve as reviewers.

Approximately seven to 10 review panels, equivalent to NIH study sections, will be organized each year, with the exact number and topical clustering of panels determined according to the number and topical areas of the proposals received. Panel management will be conducted by the four NSF directorates, with the majority conducted by CISE. Co-review across clusters, divisions and directorates will be performed where appropriate. SROs from the CSR at the NIH will be assigned to work cooperatively with NSF staff on each proposal panel. Together, they will have the responsibility to work out the details of the review process such that all agencies' needs are met. Before the review panel meetings, the representatives from the NIH SROs will work together with the NSF staff to prepare written instructions for the reviewers and to develop and implement an NIH-like scoring system (1-9) for NIH use on proposal panels. The representatives of all participating

NIH Institutes and Centers will also be invited to attend the review meetings to ensure that this review is conducted in a manner that is consistent with the agreements between NSF and the NIH.

Reviewers will be asked to evaluate proposals using two National Science Board approved merit review criteria and, if applicable, additional program specific criteria. A summary rating and accompanying narrative will generally be completed and submitted by each reviewer and/or panel. The Program Officer assigned to manage the proposal's review will consider the advice of reviewers and will formulate a recommendation.

NSF:

After scientific, technical and programmatic review and consideration of appropriate factors, the NSF Program Officer recommends to the cognizant Division Director whether the proposal should be declined or recommended for award. NSF is striving to be able to tell applicants whether their proposals have been declined or recommended for funding within six months. The time interval begins on the deadline or target date, or receipt date, whichever is later. The interval ends when the Division Director accepts the Program Officer's recommendation.

A summary rating and accompanying narrative will be completed and submitted by each reviewer. In all cases, reviews are treated as confidential documents. Verbatim copies of reviews, excluding the names of the reviewers, are sent to the Principal Investigator/Project Director by the Program Officer. In addition, the proposer will receive an explanation of the decision to award or decline funding.

In all cases, after programmatic approval has been obtained, the proposals recommended for funding will be forwarded to the Division of Grants and Agreements for review of business, financial, and policy implications and the processing and issuance of a grant or other agreement. Proposers are cautioned that only a Grants and Agreements Officer may make commitments, obligations or awards on behalf of NSF or authorize the expenditure of funds. No commitment on the part of NSF should be inferred from technical or budgetary discussions with a NSF Program Officer. A Principal Investigator or organization that makes financial or personnel commitments in the absence of a grant or cooperative agreement signed by the NSF Grants and Agreements Officer does so at their own risk.

NIH:

For those proposals that are selected for potential funding by participating NIH Institutes, the PI will be required to resubmit the proposal in an NIH-approved format. Applicants must then complete the submission process and track the status of the application in the eRA Commons, NIH's electronic system for grants administration. PIs invited to resubmit to NIH will receive further information on this process from the NIH.

Consistent with the NIH Policy for Data Management and Sharing, when data management and sharing is applicable to the award, recipients will be required to adhere to the Data Management and Sharing requirements as outlined in the [NIH Grants Policy Statement](#). Upon the approval of a Data Management and Sharing Plan, it is required for recipients to implement the plan as described. All applicants planning research (funded by NIH) that results in the generation of scientific data are required to comply with the instructions for the NIH Data Management and Sharing Plan. All applications, regardless of the amount of direct costs requested for any one year, must address a Data Management and Sharing Plan.

An applicant will not be allowed to increase the proposed budget or change the scientific content of the application in the reformatted submission to the NIH. Indirect costs on any foreign subawards/subcontracts will be limited to eight (8) percent. Applicants will be expected to utilize the Multiple Principal Investigator option at the NIH (https://grants.nih.gov/grants/multi_PI/) as appropriate.

To fulfill NIH's need for a list of participating reviewers for Summary Statements without disclosing the specific reviewers of each proposal, NSF will provide an aggregated list of the full set of reviewers for the SCH program to CSR.

Following the NSF peer review, recommended applications that have been resubmitted to the NIH are required to go to second level review by the Advisory Council or Advisory Board of the awarding Institute or Center. The following will be considered in making funding decisions:

- Scientific and technical merit of the proposed project as determined by scientific peer review.
- Availability of funds.
- Relevance of the proposed project to program priorities.
- Adequacy of data management and sharing plans.

Subsequent grant administration procedures for NIH awardees, including those related to New and Early Stage Investigators (<https://grants.nih.gov/policy/early-investigators/index.htm>), will be in accordance with the policies of NIH. Applications selected for NIH funding will use the NIH R or U funding mechanisms. At the end of the project period, renewal applications for projects funded by the NIH are expected to be submitted directly to the NIH as Renewal Applications.

VII. Award Administration Information

A. Notification of the Award

Notification of the award is made to *the submitting organization* by an NSF Grants and Agreements Officer. Organizations whose proposals are declined will be advised as promptly as possible by the cognizant NSF Program administering the program. Verbatim copies of reviews, not including the identity of the reviewer, will be provided automatically to the Principal Investigator. (See Section VI.B. for additional information on the review process.)

B. Award Conditions

An NSF award consists of: (1) the award notice, which includes any special provisions applicable to the award and any numbered amendments thereto; (2) the budget, which indicates the amounts, by categories of expense, on which NSF has based its support (or otherwise communicates any specific approvals or disapprovals of proposed expenditures); (3) the proposal referenced in the award notice; (4) the applicable award conditions, such as Grant General Conditions (GC-1)*; and (5) any announcement or other NSF issuance that may be incorporated by reference in the award notice. Cooperative agreements also are administered in accordance with NSF Cooperative Agreement Financial and Administrative Terms and Conditions (CA-FATC) and the applicable Programmatic Terms and Conditions. NSF awards are electronically signed by an NSF Grants and Agreements Officer and transmitted electronically to the organization via e-mail.

*These documents may be accessed electronically on NSF's Website at https://www.nsf.gov/awards/managing/award_conditions.jsp?org=NSF. Paper copies may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov.

More comprehensive information on NSF Award Conditions and other important information on the administration of NSF awards is contained in the *NSF Proposal & Award Policies & Procedures Guide* (PAPPG) Chapter VII, available electronically on the NSF Website at https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.

Administrative and National Policy Requirements

Build America, Buy America

As expressed in Executive Order 14005, [Ensuring the Future is Made in All of America by All of America's Workers](#) (86 FR 7475), it is the policy of the executive branch to use terms and conditions of Federal financial assistance awards to maximize, consistent with law, the use of goods, products, and materials produced in, and services offered in, the United States.

Consistent with the requirements of the Build America, Buy America Act (Pub. L. 117-58, Division G, Title IX, Subtitle A, November 15, 2021), no funding made available through this funding opportunity may be obligated for infrastructure projects under an award unless all iron, steel, manufactured products, and construction materials used in the project are produced in the United States. For additional information, visit NSF's [Build America, Buy America](#) webpage.

Special Award Conditions:

Attribution of support in publications must acknowledge the joint program, as well as the funding organization and award number, by including the phrase, "as part of the NSF/NIH Smart Health and Biomedical Research in the Era of Artificial Intelligence and Advanced Data Science Program."

NIH-Specific Award Conditions: Grants made by NSF will be subject to NSF's award conditions. Grants made by NIH will be subject to NIH's award conditions (see <http://grants.nih.gov/grants/policy/awardconditions.htm>).

C. Reporting Requirements

For all multi-year grants (including both standard and continuing grants), the Principal Investigator must submit an annual project report to the cognizant Program Officer no later than 90 days prior to the end of the current budget period. (Some programs or awards require submission of more frequent project reports). No later than 120 days following expiration of a grant, the PI also is required to submit a final annual project report, and a project outcomes report for the general public.

Failure to provide the required annual or final annual project reports, or the project outcomes report, will delay NSF review and processing of any future funding increments as well as any pending proposals for all identified PIs and co-PIs on a given award. PIs should examine the formats of the required reports in advance to assure availability of required data.

PIs are required to use NSF's electronic project-reporting system, available through Research.gov, for preparation and submission of annual and final annual project reports. Such reports provide information on accomplishments, project participants (individual and organizational), publications, and other specific products and impacts of the project. Submission of the report via Research.gov constitutes certification by the PI that the contents of the report are accurate and complete. The project outcomes report also must be prepared and submitted using Research.gov. This report serves as a brief summary, prepared specifically for the public, of the nature and outcomes of the project. This report will be posted on the NSF website exactly as it is submitted by the PI.

More comprehensive information on NSF Reporting Requirements and other important information on the administration of NSF awards is contained in the *NSF Proposal & Award Policies & Procedures Guide* (PAPPG) Chapter VII, available electronically on the NSF Website at https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.

NSF:

Additional data may be required for NSF sponsored Cooperative Agreements.

Proposals which are initially funded by NSF at a level of \$300,000 of total costs per year up to four years will be evaluated based on the proposed work plan by teams of experts periodically through the term of the project to determine performance levels. All publications, reports, data and other output from all awards must be prepared in digital format and meet general requirements for storage, indexing, searching and retrieval.

NIH:

Contact the cognizant organization program officer for additional information.

VIII. Agency Contacts

Please note that the program contact information is current at the time of publishing. See program website for any updates to the points of contact.

General inquiries regarding this program should be made to:

- Goli Yamini, Directorate for Computer and Information Science and Engineering, Division of Information and Intelligent Systems, telephone: 703-292-5367, email: sch-correspondence@nsf.gov
- Shivani Sharma, Directorate for Engineering, Division of Civil, Mechanical and Manufacturing Innovation, telephone: 703-292-4204, email: sch-correspondence@nsf.gov

- Christopher C. Yang, Directorate for Computer and Information Science and Engineering, Division of Information and Intelligent Systems, telephone: 703-292-8111, email: sch-correspondence@nsf.gov
- James E. Fowler, Computer and Information Science and Engineering, Computing and Communication Foundations, telephone: 703-292-8910, email: sch-correspondence@nsf.gov
- Sylvia Spengler, Directorate for Computer and Information Science and Engineering, Division of Information and Intelligent Systems, telephone: 703-292-8930, email: sch-correspondence@nsf.gov
- Betty K. Tuller, Social, Behavioral and Economic Sciences Directorate, Division of Behavioral and Cognitive Sciences, telephone: 703-292-7238, email: sch-correspondence@nsf.gov
- Oleg V. Sokolsky, Program Contact, telephone: (703) 292-4760, email: sch-correspondence@nsf.gov
- Dana Wolff-Hughes, National Cancer Institute (NCI), telephone: 240-620-0673, email: dana.wolff@nih.gov
- James Gao, National Eye Institute (NEI), NIH, telephone: 301-594-6074, email: james.gao@nih.gov
- Emrin Horgusluoglu, National Center for Complementary and Integrative Health (NCCIH), telephone: 240-383-5302, email: emrin.horgusluoglu-moloch@nih.gov
- Aron Marquitz, National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS), telephone: 301-435-1240, email: aron.marquitz@nih.gov
- Noffisat Oki, National Institute of Dental and Craniofacial Research (NIDCR), telephone: 301-402-6778, email: noffisat.oki@nih.gov
- Susan Wright, National Institute on Drug Abuse (NIDA), NIH, telephone: 301-402-6683, email: Susan.wright@nih.gov
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- Xujing Wang, National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), NIH, telephone: 301-451-2682, email: xujing.wang@nih.gov
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- Samantha Calabrese, Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD), NIH, telephone: 301-827-7568, email: samantha.calabrese@nih.gov

For questions related to the use of NSF systems contact:

- NSF Help Desk: 1-800-381-1532
- Research.gov Help Desk e-mail: rgov@nsf.gov

For questions relating to Grants.gov contact:

- Grants.gov Contact Center: If the Authorized Organizational Representatives (AOR) has not received a confirmation message from Grants.gov within 48 hours of submission of application, please contact via telephone: 1-800-518-4726; e-mail: support@grants.gov.

IX. Other Information

The NSF website provides the most comprehensive source of information on NSF Directorates (including contact information), programs and funding opportunities. Use of this website by potential proposers is strongly encouraged. In addition, "NSF Update" is an information-delivery system designed to keep potential proposers and other interested parties apprised of new NSF funding opportunities and publications, important changes in proposal and award policies and procedures, and upcoming NSF [Grants Conferences](#). Subscribers are informed through e-mail or the user's Web browser each time new publications are issued that match their identified interests. "NSF Update" also is available on [NSF's website](#).

Grants.gov provides an additional electronic capability to search for Federal government-wide grant opportunities. NSF funding opportunities may be accessed via this mechanism. Further information on Grants.gov may be obtained at <https://www.grants.gov>.

Main Websites for the Participating Agencies:

NATIONAL SCIENCE FOUNDATION

<https://www.nsf.gov>

NATIONAL INSTITUTES OF HEALTH

<https://www.nih.gov/>

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One or more collaborative webinar briefings with question and answer functionality will be held prior to the first submission deadline date. Schedules will be posted on the sponsor solicitation web sites.

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